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FASTENING TOOL AND FASTENING DEVICE

Abstract

A fastening tool to mount a plate-shaped photovoltaic module to a mounting target object, includes: a support member arranged on the mounting target object and including a pair of vertical wall portions and a pair of mounting portions formed at upper ends of the vertical wall portions and mounting the photovoltaic module thereon; a pressing member arranged between the pair of vertical wall portions and including a base portion fastened to the mounting target object by a fastening member via the support member, and a regulating portion provided at the base portion and regulating movement of the photovoltaic module in a direction away from the mounting target object; and a pair of claws deforming by an external force and formed at the vertical wall portions, tilted with respect to the vertical wall portions, and having a distance between tips smaller than a width of the base portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation application of PCT Application No. PCT/JP2022/044254, filed Nov. 30, 2022, the entire contents of which are incorporated herein by reference.

FIELD
[0002] The present invention relates to a fastening tool and a fastening device for fastening photovoltaic modules to an installation position.
BACKGROUND
[0003] Photovoltaic power generation systems for generating power with multiple photovoltaic modules fastened, for example, to a roof of a house have been known. As a technique for fastening the photovoltaic modules to the roof, Jpn. Pat. Appln. KOKAI Publication No. 2018-109322 discloses a technique of securing a base member between two photovoltaic modules adjacent to each other on the roof and fastening to this base member a pressing member for pressing edge portions of the two adjacent photovoltaic modules with bolts and nuts.
[0004] The fastening by the bolts and the nuts, however, has the following issues. That is, in order to confirm that the bolts and the nuts have not been forgotten to be fastened, for example, it is conceivable that a worker would manually check the looseness of the bolts and the nuts, but such manual checking work is inefficient.
SUMMARY
[0005] According to one aspect of the embodiment, the fastening tool is a fastening tool to mount a plate-shaped photovoltaic module to a mounting target object, and it includes: a support member including a pair of vertical wall portions and a pair of mounting portions formed at upper ends of the pair of vertical wall portions and configured to mount the photovoltaic module thereon, the support member being arranged on the mounting target object; a pressing member including a base portion fastened to the mounting target object by a fastening member via the support member, and a regulating provided at the base portion and configured to regulate movement of the photovoltaic module in a direction away from the mounting target object, the pressing member being arranged between the pair of vertical wall portions; and a pair of claws configured to deform by an external force, the pair of claws being formed at the pair of vertical wall portions, being tilted with respect to the vertical wall portions, and having a distance between tips smaller than a width of the base portion. The pair of claws are configured to, prior to the fastening of the base portion to the mounting target object by the fastening member, abut on the base portion, and tilt the pressing member with respect to a posture of the regulating portion regulating movement of the photovoltaic module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] FIG. **1** is an exploded perspective view showing a photovoltaic power generation system using a fastening device according to a first embodiment of the present invention.
[0007] FIG. **2** is a cross-sectional view showing the photovoltaic power generation system.
[0008] FIG. **3** is a perspective view showing the fastening device.
[0009] FIG. **4** is a side view showing a configuration of the photovoltaic power generation system

in a state before an eaves-side photovoltaic module is fastened.

[0010] FIG. **5** is a side view showing the configuration of the photovoltaic power generation system in a state where the eaves-side photovoltaic module is fastened.

[0011] FIG. **6** is a side view showing the configuration of the photovoltaic power generation system in a state where a ridge-side photovoltaic module is inserted.

[0012] FIG. **7** is a front view showing a configuration of a support member used in a fastening tool of the fastening device.

[0013] FIG. **8** is a plan view showing a structure of the support member.

[0014] FIG. **9** is a side view showing the structure of the support member.

[0015] FIG. **10** is a front view showing a dimensional structure of the support member and a pressing member of the fastening tool.

DETAILED DESCRIPTION

[0016] A fastening device **3** and a photovoltaic power generation system **1** using the fastening device **3** according to an embodiment of the present invention will be described with reference to FIGS. **1** to **10**. FIG. **1** is an exploded perspective view showing the photovoltaic power generation system **1**. FIG. **2** is a cross-sectional view showing the photovoltaic power generation system **1**. FIG. **3** is a perspective view showing the fastening device **3**. FIG. **4** is a side view showing a configuration of the photovoltaic power generation system **1** in a state before an eaves-side photovoltaic module **2** is fastened. FIG. **5** is a side view showing the configuration of the photovoltaic power generation system **1** in a state where the eaves-side photovoltaic module **2** is fastened. FIG. **6** is a side view showing the configuration of the photovoltaic power generation system **1** in a state where a ridge-side photovoltaic module **2** is inserted. FIG. **7** is a front view showing the configuration of a support member **22** used in a fastening tool **6** of the fastening device **3**. FIG. **8** is a plan view showing a structure of the support member **22**. FIG. **9** is a side view showing the structure of the support member **22**. FIG. **10** is a front view showing a dimensional structure of the support member **22** and a pressing member **23** of the fastening tool **6**. In FIGS. **2** and **4** to **6**, the photovoltaic power generation system **1** or the fastening device **3** is shown in a posture in which the right side is the ridge side and the left side is the eaves side.

[0017] As illustrated in FIGS. **1** and **2**, the photovoltaic power generation system **1** is installed on a roof **200** of a house which is an example of an installation position. As illustrated in FIG. **2**, the roof **200** includes, for example, a plurality of roofing members **201**. The roofing members **201** are, for example, slates. The roofing members **201** are arranged on an upper surface **203** of a roof board **202**. The ridge-side end portion of the roofing member **201** is placed on the roof board **202**, and the eaves-side end portion of the roofing member **201** is placed on another roofing member **201** on the eaves side of this roofing member **201**. As shown in FIG. **2**, a first direction V**1**, which is a direction of an upper surface **204** of the roofing member **201** tilted with respect to an up-down direction, intersects a second direction V**2**, which is a direction of the upper surface **203** of the roof board **202** tilted with respect to the up-down direction.

[0018] As shown in FIG. **1**, the photovoltaic power generation system **1** includes the plurality of photovoltaic modules **2**, the plurality of fastening devices **3**, and a plurality of decorative plates **5**. The photovoltaic power generation system **1** may be configured to use a plurality of eaves-side fastening devices different from the fastening device **3** as the fastening device for fastening the eaves side of the photovoltaic module **2** arranged closest to the eaves side of the plurality of arranged photovoltaic modules **2**.

[0019] The plurality of photovoltaic modules **2** are attached to the roof **200** by the plurality of fastening devices **3** or by the plurality of fastening devices **3** and the plurality of eaves-side fastening devices. The plurality of photovoltaic modules **2** are arranged in both a third direction V**3**, which is parallel to a direction from the ridge toward the eaves, and a fourth direction V**4**, which is orthogonal to the third direction V**3**, for example. The number of the photovoltaic modules **2** is suitably determined in accordance with the size of the installation position, a targeted output, and

the like.
[0020] The photovoltaic module **2** is configured in a plate shape. The photovoltaic module **2** includes, for example, a solar panel **10** and a frame **11**. The solar panel **10** is configured by integrally attaching together a plurality of cells **12** that are constituted of solar cell elements and forming them into a plate. The solar panel **10** is configured in a rectangular plate shape, for example.
[0021] As shown in FIG. **2**, the frame **11** is provided at an edge portion of the solar panel **10**. In FIG. **2**, the cross section of the photovoltaic module **2** is illustrated. The frame **11** includes a base part formed of a conductive metal material and a protective layer provided on a surface of the base part for corrosion prevention. The protective layer is formed by, for example, plating the base part.
[0022] The frame **11** includes, for example, an upper surface portion **13**, a side surface portion **14**, and a lower surface portion **15**. The upper surface portion **13** is formed to surround an edge portion of an upper surface **16**, an edge portion of a lower surface **17**, and a side surface **18** of the solar panel **10**. The side surface portion **14** extends downward from the upper surface portion **13**. The lower surface portion **15** extends from the lower end of the side surface portion **14** toward the inside of the frame **11**.
[0023] As shown in FIGS. **1** and **2**, the fastening device **3** is provided, for example, between two photovoltaic modules **2** adjacent in the third direction V**3**. The fastening device **3** fastens the two adjacent photovoltaic modules **2** to the roof **200**.
[0024] As shown in FIGS. **2** and **3**, the fastening device **3** includes, for example, a base member **20**, a fastening member **21** for fastening the base member **20** to the roof **200**, a fastening tool **6**, and a bolt **24** and a nut **25** that serve as a fastening member for fastening the fastening tool **6** to the base member **20**.
[0025] The base member **20** is an example of a mounting target object on which the photovoltaic module **2** is mounted. The base member **20** is formed in a shape elongated in one direction, for example, the third direction V**3**, when fastened to the roof **200**. The length of the base member **20** in the longitudinal direction is determined to be shorter than that of the photovoltaic module **2** in the third direction V**3**. The roof board **202** is an example of a base part of the roof **200** on which the roofing members **201** are provided.
[0026] The base member **20** includes, for example, a base portion **30**, a pair of vertical wall portions **34**, and a pair of flanges **31**, as shown in FIGS. **2** and **3**.
[0027] The upper surface of the base portion **30** is formed into a flat surface. The upper surface of the base portion **30** is formed, for example, into a plane that becomes parallel to the upper surface **203** of the roof board **202**, which is an example of a base part, in a state where the base member **20** is fastened to the roof **200**. The base portion **30** is formed into a plate shape elongated in one direction. The base portion **30** includes a groove **35** and a regulating portion **36**.
[0028] The groove **35** pierces through the base portion **30** and extends along the longitudinal direction of the base portion **30**. The groove **35** is provided, for example, in the center in the width direction of the base portion **30**, extending from the vicinity of one end of the base portion **30** to the vicinity of the other end in the longitudinal direction. In the groove **35**, a screw portion **24***c* of the bolt **24**, which will be described later, can be arranged in a movable manner in the longitudinal direction of the groove **35**.
[0029] The groove **35** includes a first portion **37** and a second portion **38**. The first portion **37** forms, for example, one end portion of the groove **35**. The length and width of the first portion **37** along the longitudinal direction of the groove **35** are set to dimensions that allow a head portion **24***a*, which will be described later, of the bolt **24** to pass through. The first portion **37** is provided at the eaves-side end portion of the groove **35** in a state where the base member **20** is fastened to the roof **200**.
[0030] The second portion **38** forms the other end side of the groove **35** with respect to the first portion **37**. The length and width of the second portion **38** along the longitudinal direction of the

groove **35** are set to dimensions that allow the screw portion **24***c* to be arranged and that do not allow the head portion **24***a* to pass through. In the present embodiment, for example, the width of the second portion **38** is set to be sufficient for a base portion **24***b*, which will be described later, of the bolt **24** to move through. In the second portion **38** having the edges extending along the longitudinal direction with respect to the width direction, one of these edges is linearly continuous with the edge of the first portion **37**.

[0031] The regulating portion **36** is configured to regulate the rotation of the bolt **24**. The regulating portion **36** is provided on, for example, the lower surface of the base portion **30**. The regulating portion **36** includes, for example, a pair of vertical wall portions **36***a* at the two edges of the second portion **38** along the longitudinal direction. The distance between the pair of vertical wall portions **36***a* is set to such a distance that the base portion **24***b* of the bolt **24** can be arranged in between and that the rotation of the bolt **24** can be regulated with the inner surfaces of the pair of vertical wall portions **36***a* abutting against the outer surface of the base portion **24***b*.

[0032] In another example, the regulating portion **36** may be configured by the inner surface of the groove **35**. For instance, in the case where the thickness of the base portion **30** is sufficient to receive at least part of the base portion **24***b* in the groove **35** with the bolt **24** and the nut **25** engaged with each other, the rotation of the bolt **24** can be regulated with the inner surface of the groove **35** abutting on the base portion **24***b*.

[0033] The pair of vertical wall portions **34** are formed, for example, at the edges of the base portion **30** along the longitudinal direction. The length of the pair of vertical wall portions **34** in a vertical direction is set to a dimension such that the upper surface of the base portion **30** is a plane parallel to the upper surface **203** of the roof board **202** in a state where the base member **20** is fastened to the roof **200**. In the example of the present embodiment, the vertical length of the vertical wall portion **34** is set to a dimension that gradually increases from one end on the eaves side, that is, one end on the first portion **37** side, toward the other end side.

[0034] The pair of flanges **31** are formed at the lower end edges of the outer surfaces of the pair of vertical wall portions **34**. The pair of flanges **31** are formed in a rectangular plate shape, for example. One or more holes **32** are formed in each flange **31**, through which a fastening member **21** is inserted. The hole **32** is formed, for example, on each of one end side and the other end side of the flanges **31** in the longitudinal direction.

[0035] The fastening member **21** is, for example, a screw. The fastening member **21** is inserted into the hole **32** and engaged to the roofing member **201** and the roof board **202**, thus fastening the base member **20** to the roofing member **201** and the roof board **202**.

[0036] The fastening tool **6** includes a support member **22** mounted on the base member **20**, and a pressing member **23** to press the upper surface portion **13** of the frame **11**, which is an example of the upper surface **16** of the photovoltaic module **2**.

[0037] The support member **22** is mounted on the upper surface of the base portion **30**, as shown in FIGS. **2** to **6**. The support member **22** includes a support portion that supports a part of the lower surface of the photovoltaic module **2**, which is an example of the main surface of the photovoltaic module **2** on the base member **20** side, for example, a part of the edge portion. That is, as shown in FIGS. **2** and **4** to **6**, the support member **22** carries the photovoltaic module **2** and is arranged on the base member **20**. As shown in FIGS. **2** to **9**, the support member **22** includes, for example, a support member base portion **41**, a pair of support portions **42**, and a pair of claws **44**. The support member **22** is formed of a conductive material. The support member **22** is formed of, for example, aluminum, iron, or a resin. Preferably, the support member **22** is made of a material that can be subjected to press working, and, for example, is preferably formed of a steel material containing iron as a main material.

[0038] The support member base portion **41** is mounted on the upper surface of the base portion **30**. The support member base portion **41** is formed into a rectangular plate shape, for example. As a specific example, the support member base portion **41** is formed into a rectangular plate shape which is short in the width direction of the base member **20**, that is, long in the longitudinal direction of the base member **20**. The length (width) of the support member base portion **41** in the lateral direction is greater than the width of the base portion **30**. The support member base portion **41** has a hole **45** in which the threaded portion **24***c* of the bolt **24** is arranged. The support member base portion **41** has a guided portion **46** that is arranged in the groove **35** of the base member **20** and guides the support member **22** to move along the groove **35**.

[0039] The guided portion **46** is set to have a width smaller than the width of the groove **35** in a direction orthogonal to the longitudinal direction of the groove **35**, for example, and is formed to be movable in the groove **35** along the longitudinal direction of the groove **35**. The guided portion **46** is formed by, for example, bending a part of the support member base portion **41** toward the groove **35** side which is the lower side.

[0040] The support portion **42** supports two photovoltaic modules **2** on the upper surface. More specifically, the support portion **42** supports a part of the edge portion of the lower surface of each of the two adjacent photovoltaic modules **2** aligned in the third direction V**3**. The pair of support portions **42** are formed at both ends of the support member base portion **41** in the lateral direction (the fourth direction V**4**). The support portion **42** includes, for example, a vertical wall portion **47** and a mounting portion **48**.

[0041] The vertical wall portion **47** is formed into a plate shape standing upright from the support member base portion **41**. The mounting portion **48** is provided on an upper end of the vertical wall portion **47**. A hole **47***a* is formed in the vertical wall portion **47**. For example, a ground wire is fixed in the hole **47***a* formed in one of the pair of vertical wall portions **47**, and for example, wiring of the photovoltaic module **2** is fixed in the hole **47***a* formed in the other of the pair of vertical wall portions **47**. Therefore, for example, as shown in FIGS. **7** to **9**, the hole **47***a* in which the ground wire is fixed is formed to have a smaller diameter than the hole **47***a* in which the wiring is fixed. Further, for example, the hole **47***a* is arranged on the ridge-side of the vertical wall portion **47**. The number, shape, size, and the like of the holes **47***a* can be appropriately changed depending on various wires to be fixed and the intended use.

[0042] A part of the edge portion of the lower surface of the photovoltaic module **2** is mounted on the mounting portion **48**. The mounting portion **48** is formed into a rectangular plate shape that extends in a direction opposite to the direction in which the pair of vertical wall portions **47** face each other (the fourth direction V**4**) and is partially cut out, for example. The mounting portion **48** is cut out in the center side in the longitudinal direction (direction orthogonal to the facing direction of the pair of vertical wall portions **47**, third direction V**3**) of the support member base portion **41**, and has a first protrusion **48***a* provided at the cut-out portion and a second protrusion **48***b* provided closer to the eaves side than the first protrusion **48***a* in the third direction V**3**.

[0043] The first protrusion **48***a* is formed to have, for example, a sharp tip so that the protection layer on the surface of the side surface portion **14** of the frame **11** of the ridge-side photovoltaic module **2** (one photovoltaic module) can be damaged by the tip. As a specific example, in the first protrusion **48***a*, for example, a part of the mounting portion **48** is formed in a triangular shape, for example, in a two right-angled triangle or an isosceles sided shape of a triangle. The first protrusion **48***a* is tilted with respect to each of the upper surface of the mounting portion **48** and a direction intersecting a direction orthogonal to the upper surface of the mounting portion **48**, and extends toward the ridge-side. The first protrusion **48***a* is formed by, for example, cutting out the central portion of the mounting portion **48** into two sides of a triangle and bending the central portion by press working. In the case where the photovoltaic module **2** supported on the ridge-side of the support member **22** is mounted on the mounting portion **48**, the first protrusion **48***a* breaks through the protection layer on the eaves-side side surface of the frame **11** and abuts on the base part of the frame **11**.

[0044] The base end portion of the first protrusion **48***a* continuous with the mounting portion **48** is provided at a portion where the photovoltaic module **2** is mounted on the eaves side of the mounting portion **48** or a portion facing the first regulating portion **62** of the pressing member **23**. The tip of the first protrusion **48**a is arranged above the upper surface of the mounting portion **48**, and up to a position where the side surface portion **14** of the frame **11** of the ridge-side photovoltaic module **2** of the mounting portion **48** is arranged or up to a position slightly closer to the ridge side than the position where the side surface portion **14** of the frame **11** of the ridge-side photovoltaic module **2** is arranged.

[0045] The second protrusion **48**b is formed to have, for example, a sharp tip so that the protection layer on the surface of the lower surface portion **15** of the frame **11** of the eaves-side photovoltaic module **2** (the other photovoltaic module) can be damaged by the tip. As a specific example, the second protrusion **48**b is formed by, for example, a part of the mounting portion **48** being formed in a triangular shape, for example, in a two sided shape of a right-angled triangle or an isosceles triangle. The second protrusion **48**b is, for example, orthogonal to the upper surface of the mounting portion **48** and projects upward. The second protrusion **48**b is formed by, for example, cutting out a part of the mounting portion **48** into two sides of a triangle and bending the cutout by press working. In the case where the photovoltaic module **2** supported on the eaves side of the support member **22** is mounted on the mounting portion **48**, the second protrusion **48**b breaks through the protection layer on the surface of the lower surface of the frame **11** and abuts on the base part of the frame **11**.

[0046] The pair of claws **44** are provided on the pair of support portions **42**. The pair of claws **44** are provided to face each other in the width direction of the support member **22** (the facing direction of the pair of vertical wall portions **47**, the fourth direction V**4**). The claw **44** is integrally provided on the support portion **42**, and a tip thereof is arranged closer to the center of the support member **22** than the inner surface of the vertical wall portion **47**. The distance of the gap in the direction in which the pair of claws **44** face each other (the width direction of the support member **22**, the fourth direction V**4**) gradually decreases from the upper side (the mounting portion **48**) toward the lower side (the support member base portion **41**). That is, the claw **44** is tilted at a predetermined angle with respect to the inner surface of the vertical wall portion **47** so as to be tilted from the upper side to the lower side. In other words, the distance between the pair of claws **44** in the facing direction is the same as the distance between the pair of vertical wall portions **47** at the upper end of the claw **44**, and is smaller than the distance between the pair of vertical wall portions **47** at the lower end of the claw **44**.

[0047] As shown in FIG. **10**, the distance W**1** between the lower ends of the pair of claws **44** is smaller than the width W**2** of the base portion **60**, which will be described later, of the pressing member **23** in the fourth direction V**4**, so that the pressing member **23** can be regulated from moving downward beyond the claws **44**. The pair of claws **44** are formed to be elastically deformable by an external force.

[0048] For example, the upper end of the claw **44** is continuous with the vertical wall portion **47**. As a specific example, the claw **44** is formed integrally with the vertical wall portion **47** by punching out a part of the vertical wall portion **47** shown in FIG. **9** into a rectangular shape except for one side, or a U-shape, a V-shape, or the like, so that the upper end is continuous with the vertical wall portion **47**, and plastically deforming the punched out part by bending so as to form a predetermined angle smaller than 90° with respect to the vertical wall portion **47**. With these pair of claws **44**, by the pair of claws **44** abutting on the base portion **60**, which will be described later, of the pressing member **23**, the pressing member **23** is provided at a position tilted with respect to the support member base portion **41**. As a specific example, the pair of claws **44** are provided to the center side in the longitudinal direction (third direction V**3**) of the vertical wall portions **47**.

[0049] As shown in FIGS. **2** to **6**, the pressing member **23** is mounted on the support member base portion **41** of the support member **22**. The pressing member **23** is configured to be able to regulate movement of the photovoltaic module **2**. The pressing member **23** includes, for example, the base portion **60** and a regulating portion **61**. The pressing member **23** is formed of, for example, iron, aluminum, or stainless steel.

[0050] The base portion **60** configures, for example, the lower end of the pressing member **23**, and is arranged on the pair of claws **44** when arranged in the support member **22**, and is moved onto the support member base portion **41** against the pair of claws **44** by being engaged by the bolt **24** and the nut **25**, and is thus arranged on the support member base portion **41**. The base portion **60** is formed in a rectangular plate shape that is long in a direction (longitudinal direction, third direction V**3**) orthogonal to the width direction of the base member **20**, for example. The base portion **60** has a width W**2** (length in the lateral direction of the base portion **60**) that is smaller than the distance between the upper ends of the pair of claws **44** in the facing direction and greater than the distance W**1** between the lower ends of the pair of claws **44** in the facing direction. The base portion **60** has a hole **60***a* formed in a shape that allows the screw portion **24***c* of the bolt **24** to be inserted therethrough but does not allow the head portion **24***a* of the bolt **24** and the nut **25** to be inserted therethrough. In a specific example, the hole **60***a* is formed as a round hole having a radius greater than that of the screw portion **24***c* and smaller than the minimum width of the head portion **24***a*. The hole **60***a* has a dimension that allows the pressing member **23** to be moved between a first state P**1** and a second state P**2**, which will be described later, with the screw portion **24***c* arranged.

[0051] The regulating portion **61** is provided in the base portion **60** and is configured to be able to regulate movement of the photovoltaic module **2**. The regulating portion **61** includes, for example, a first regulating portion **62** and a second regulating portion **63**.

[0052] The first regulating portion **62** is formed in a shape that stands upward from one end portion of the base portion **60**, in other words, the eaves side of the base portion **60**, specifically, the edge on the eaves side of the upper surface of the base portion **60**. For example, the first regulating portion **62** extends from the base portion **60** in a direction of 90° with respect to the base portion **60**. The first regulating portion **62** is formed in a rectangular plate shape, for example. The ridge-side surface of the first regulating portion **62** abuts against the eaves-side side surface portion **14** of the ridge-side photovoltaic module **2**, thereby regulating the movement of the photovoltaic module **2** toward the eaves side. A ridge-side surface of the first regulating portion **62** is provided with a regulating protrusion **62***a* that faces a part of an edge portion of the lower surface of the ridge-side photovoltaic module **2**.

[0053] The regulating protrusion **62***a* is formed in a stepped shape having an upper surface at a different height position or a flat plate shape, for example. In the present embodiment, the regulating protrusion **62***a* is formed in a stepped shape descending toward the base portion **60** side in a direction away from the first regulating portion **62**. The regulating protrusion **62***a* is at least partially positioned above the upper surface of the mounting portion **48** of the support member **22** in the second state P**2** where the pressing member **23** is tilted by the pair of claws **44**, and when the ridge-side photovoltaic module **2** is inserted, regulates the movement of the photovoltaic module **2**. In the first state P**1** in which the pressing member **23** passes over the pair of claws **44** and the base portion **60** abuts on the support member base portion **41**, the regulating protrusion **62***a* is positioned below the upper surface of the mounting portion **48** of the support member **22** and is formed so that the ridge-side photovoltaic module **2** can be inserted.

[0054] The eaves-side surface of the first regulating portion **62** abuts on the ridge-side side surface portion **14** of the eaves-side photovoltaic module **2**, thereby regulating the movement of the photovoltaic module **2** toward the ridge side.

[0055] The first regulating portion **62** may abut on the side surface portion **14** of the photovoltaic module **2** with the photovoltaic module **2** attached to the base member **20**. Alternatively, the first regulating portion **62** may have a gap between the first regulating portion **62** and the side surface portion **14**, and if the photovoltaic module **2** moves toward the eaves side or the ridge side by a predetermined distance, the first regulating portion **62** may abut on the side surface portion **14**, thereby regulating the movement of the photovoltaic module **2**.

[0056] The second regulating portion **63** is formed, for example, on the upper end of the first regulating portion **62**. The second regulating portion **63** comes into contact with the upper surface portion **13** of the frame **11**, which is an exemplary main surface of the photovoltaic module **2** opposite to the base member **20**, thereby regulating the movement of the photovoltaic module **2** in a direction away from the base member **20**. The second regulating portion **63** includes, for example, a third regulating portion **65** and a fourth regulating portion **66**.

[0057] The third regulating portion **65** is formed into a plate extending from the first regulating portion **62** to the ridge side. The third regulating portion **65** faces the base portion **60** and the regulating protrusion **62***a* in the longitudinal direction of the first regulating portion **62**. As indicated by the double-dashed chain line in FIG. **2**, the distance between the third regulating portion **65** and the regulating protrusion **62***a* is set to be greater than the dimension (thickness) along the vertical direction of the frame **11** of the photovoltaic module **2**, such that the photovoltaic module **2** held in an oblique posture can be inserted between the third regulating portion **65** and the regulating protrusion **62***a* when the pressing member **23** is in a first posture P**1**. As indicated by the double-dashed chain line in FIG. **2**, the distance between the third regulating portion **65** and the regulating protrusion **62***a* is set to be greater than the thickness of the frame **11** such that the photovoltaic module **2** held in the oblique posture cannot be inserted between the third regulating portion **65** and the regulating protrusion **62***a* when the pressing member **23** is in the second posture P**2** as shown in FIG. **4**.

[0058] Here, as indicated by the double-dashed chain line in FIG. **2**, the oblique posture of the photovoltaic module **2** indicates a posture in which the longitudinal direction (main surface direction) of the photovoltaic module **2** is tilted with respect to the extending direction of the third regulating portion **65** and the regulating protrusion **62***a*, and the ridge-side end of the photovoltaic module **2** is arranged above the eaves-side end.

[0059] The third regulating portion **65** is separated from the upper surface of the photovoltaic module **2** (frame **11**) with a gap therebetween in a state where the photovoltaic module **2** is attached to the fastening tool **6**, and regulates the movement of the photovoltaic module **2** by abutting on the photovoltaic module **2** when the photovoltaic module **2** is moved upward by a predetermined distance. The third regulating portion **65** may be configured to abut on the upper surface portion **13** of the photovoltaic module **2**.

[0060] The fourth regulating portion **66** is formed into a plate extending to the eaves side. The fourth regulating portion **66** faces the support member base portion **41** of the support member **22** when the regulating portion **61** is in the first state P**1**. The distance between the fourth regulating portion **66** and the mounting portion **48** when the regulating portion **61** is in the first state P**1** is set to be equal to the thickness of the edge portion of the photovoltaic module **2**. The fourth regulating portion **66** abuts on the upper surface portion **13** of the photovoltaic module **2** in a state where the photovoltaic module **2** is attached to the fastening tool **6**, and presses the photovoltaic module **2**, thereby fastening the photovoltaic module **2**.

[0061] In the case where the bolt **24** and the nut **25** inserted through the groove **35** of the base member **20**, the hole **45** of the support member **22**, and the hole **60***a* of the base portion **60** are engaged, the pressing member **23** configured as described above presses the pair of claws **44** by the base portion **60** and deforms the pair of claws **44**, and is in the first state P**1** in which the base portion **60** abuts on the upper surface of the support member base portion **41**. Here, the first state P**1** is a state in which the regulating portion **61** can regulate the movement of the photovoltaic module **2**. In the present embodiment, the first state P**1** is a posture of the pressing member **23** in which the first regulating portion **62** can abut on the side surface portion **14** of the frame **11** of the photovoltaic module **2** and the fourth regulating portion **66** of the second regulating portion **63** abuts on the upper surface portion **13** of the frame **11** of the photovoltaic module **2** on the eaves side.

[0062] In a temporarily fixed state of the pressing member **23** and before the first state P**1**, the pressing member **23** is in the second state P**2** in which the base portion **60** is arranged on the pair of claws **44** and the first regulating portion **62** is tilted with respect to the upper surface of the support member base portion **41**. Here, the temporarily fixed state means that the pressing member **23** is arranged on the support member **22**, and the nut **25** and the bolt **24** are partially screwed, but the nut **25** and the bolt **24** are not completely engaged. In the second state P**2**, the pressing member **23** is in a posture tilted toward the ridge-side with respect to the first state P**1**. In the second state P**2**, the pressing member **23** has a greater gap between the fourth regulating portion **66** and the upper surface portion **13** of the photovoltaic module **2** than in the first state P**1**.

[0063] When the pressing member **23** moves from the second state P**2** to the first state P**1**, the pair of claws **44** having the distance W**1** smaller than the width W of the base portion **60** are pressed by the base portion **60** and elastically deformed, and the distance W**1** between the pair of claws **44** becomes greater than the width W**2** of the base portion **60**. When the base portion **60** passes over the pair of elastically deformed claws **44**, the pair of claws **44** are restored, and the distance W**1** between the pair of claws **44** becomes smaller than the width W**2** of the base portion **60**. Thus, in the case where the pressing member **23** moves from the second state P**2** to the first state P**1**, and then moves from the first state P**1** to the second state P**2**, the base portion **60** abuts on the pair of claws **44** that have been restored, and thus the movement of the pressing member **23** to the second state P**2** is regulated.

[0064] The bolt **24** and the nut **25** are an example of a fastening member that fastens the support member **22** and the pressing member **23** to the base member **20**. The bolt **24** has, for example, a head portion **24***a* and the screw portion **24***c* in which a female screw is formed. The bolt **24** and the nut **25** are engaged with the base member **20**, the support member **22**, and the pressing member **23** interposed, thereby fastening the support member **22** and the pressing member **23** to the base member **20**. The bolt **24** and the nut **25** are engaged to press the pressing member **23** downward and elastically deform the pair of claws **44**, thereby moving the pressing member **23** from the second state P**2** to the first state P**1** against the pair of claws **44**.

[0065] The bolt **24** has an abutting portion which abuts on the regulating portion **36** to regulate the rotation of the bolt **24**. The abutting portion is provided, for example, between the head portion **24***a* and the threaded portion **24***c* of the bolt **24**. The bolt **24** is, for example, a square-root round-head bolt, and has a head portion **24***a*, a base portion **24***b* as an example of an abutting portion, and the screw portion **24***c*. The base portion **24***b* is formed in a polygonal shape, for example, a rectangular shape, or has a pair of flat surfaces formed at symmetrical positions which abut on the pair of regulating portions **36**. The diameter of the circumscribed surface of the base portion **24***b* is greater than the width of the groove **35**, and the width of two surfaces that are paired (at symmetrical positions) among the plurality of flat surfaces formed on the base portion **24***b* is smaller than the width of the groove **35**.

[0066] With the bolt **24**, in a posture in which the head portion **24***a* is arranged below the base portion **30** of the base member **20**, for example, the screw portion **24***c* is inserted through the groove **35** of the base member **20**, the hole **45** of the support member **22**, and the hole **60***a* of the pressing member **23**. The nut **25** is screwed into, for example, a portion of the screw portion **24***c* above the base portion **60**. The nut **25** is engaged with the bolt **24** with the support member base portion **41** of the support member **22** and the base portion **60** of the pressing member **23** interposed.

[0067] The decorative plate **5** covers the fastening tool **6** positioned closest to the eaves side. In the case where the photovoltaic power generation system **1** includes an eaves-side fastening device, in addition to the fastening device **3**, the decorative plate **5** covers the eaves-side fastening device. Here, the eaves-side fastening device fastens, for example, the photovoltaic module **2** arranged closest to the eaves side. The eaves-side fastening device is an example of a fastening device that fastens the photovoltaic module **2**. The eaves-side fastening device may be the fastening tool **6**.

Next, the operation of fastening the photovoltaic module **2** to the roof **200** using the fastening device **3** will be described. In the following description, the photovoltaic module **2** arranged on the eaves side of the fastening device **3** (the pressing member **23**) will be referred to as a first photovoltaic module **2**A, and the photovoltaic module **2** arranged on the ridge side of the fastening device **3** will be referred to as a second photovoltaic module **2**B.

[0068] First, the worker arranges the plurality of eaves-side fastening devices and the plurality of fastening devices **3** to the fastening positions on the roof **200**. At this time, the pressing member **23** of the fastening device **3** is in the second state P**2** as shown in FIG. **4**. The worker arranges the eaves-side end portion of the first photovoltaic module **2**A on the eaves-side fastening device **3** or the eaves-side fastening device, and places the ridge-side end portion of the first photovoltaic module **2**A on the mounting portion **48** of the support member **22** of the fastening device **3** on which the eaves-side end portion of the first photovoltaic module **2**A is arranged or the fastening device **3** closer to the ridge side than the eaves-side fastening device.

[0069] At this time, the worker moves the support member **22** and the pressing member **23** of the fastening device **3**, on which the ridge-side end portion of the first photovoltaic module **2**A is placed, along the groove **35**. By this operation, the worker adjusts the positions of the support member **22** and the pressing member **23** to suitable positions with respect to the photovoltaic module **2**A, for example, to positions at which the frame **11** and the first regulating portion **62** abut on each other. In the case where the first photovoltaic module **2**A is mounted on the mounting portion **48**, the second protrusion **48***b* breaks the protection layer of the frame **11** so as to abut on the base part of the frame **11**. Thus, the base part of the frame **11** and the support member **22** are electrically connected to each other.

[0070] Next, the worker engages the bolt **24** and the nut **25** inserted into the groove **35** of the base member **20**, the hole **45** of the support member **22**, and the hole **60***a* of the base portion **60**, and fastens the support member **22** and the pressing member **23** to the base member **20**. By engaging the bolt **24** and the nut **25**, the base portion **60** of the pressing member **23** presses the pair of claws **44**. When the pair of claws **44** pressed by the base portion **60** are elastically deformed, the base portion **60** rides on the pair of claws **44**. As a result, as illustrated in FIG. **5**, the pressing member **23** moves from the second state P**2** to the first state P**1**. The pressing member **23** in the first state P**1** sandwiches the eaves-side photovoltaic module **2**A between the pressing member **23** and the support member **22**, and fastens the first photovoltaic module **2**A.

[0071] Next, the worker visually recognizes the plurality of fastening devices **3** for fastening the first photovoltaic module **2**A, and checks that the bolts **24** and the nuts **25** are engaged. In the case where it is in the temporarily fixed state where the bolt **24** and the nut **25** are not fastened, as shown in FIG. **4**, because the pressing member **23** is in the second state P**2**, the first regulating portion **62** and the second regulating portion **63** are tilted with respect to the upper surface of the mounting portion **48** of the support member **22** (the upper surface of the photovoltaic module **2**), and thus the worker can easily visually recognize the temporarily fixed state.

[0072] Next, as shown by the double-dashed chain line in FIG. **2**, the worker inserts the second photovoltaic module **2**B between the third regulating portion **65** and the mounting portion **48** in a posture in which the upper surface of the second photovoltaic module **2**B is tilted with respect to the upper surface of the mounting portion **48**. At this time, the first protrusion **48***a* breaks the protection layer of the side surface portion **14** of the frame **11** of the second photovoltaic module **2**B to abut on the base part, and thus the base part of the frame **11** of the second photovoltaic module **2**B and the support member **22** become electrically conductive. Thus, the first photovoltaic module **2**A and the second photovoltaic module **2**B become electrically conductive via the support member **22**. Next, the worker lowers the ridge-side end portion of the second photovoltaic module **2**B and places the ridge-side end portion of the second photovoltaic module **2**B on the mounting portion **48** of the support member **22** facing the ridge-side end portion of the second photovoltaic module **2**B.

[0073] Next, in the same manner as the above fastening operation of the ridge-side end portion of the first photovoltaic module **2**A by the fastening device **3**, the worker fastens the ridge-side end portion of the second photovoltaic module **2**B by the fastening device **3**. The worker repeats the

above steps to perform the fastening operation of the first photovoltaic module **2**A and the second photovoltaic module **2**B using the fastening device **3**, and sequentially fastens the photovoltaic modules **2** to the roof **200** from the eaves side toward the ridge side.
[0074] The plurality of photovoltaic modules **2** aligned in one row in the third direction V**3** become electrically conductive via the support members **22** of the plurality of fastening devices **3** and the plurality of frames **11**. The worker attaches the ground wire to the support member **22** of the fastening device **3** closest to the ridge side, for example, and grounds all the plurality of photovoltaic modules **2** aligned in a row from the eaves side to the ridge side, which are electrically conductive by the support members **22** of the plurality of fastening devices **3**. The ground wire is electrically connected and fastened to, for example, a hole **47***a* for fixing the ground wire formed in one vertical wall portion **47** of the support member **22**. In this manner, the fastening operation of the plurality of photovoltaic modules **2** to the roof **200** using the plurality of fastening devices **3** is completed.
[0075] In the fastening device **3** configured as described above, the pressing member **23** is in the second state P**2**, if the bolt **24** and the nut **25** are not engaged in the temporarily fixed state, with the pressing member **23** tilted by the pair of claws **44**. Therefore, the worker can visually check that the pressing member **23** is in the second state P**2**, and thus can recognize that the bolt **24** and the nut **25** are not engaged. This can improve efficiency of the checking work, by the worker being able to easily check that the bolt **24** and the nut **25** are not left unengaged, that is, whether the fastening tool **6** is left unfastened. In particular, since a plurality of the fastening devices **3** are used for fastening one photovoltaic module **2**, it takes time to check whether the photovoltaic module **2** has been forgotten to be fastened when the number of the photovoltaic modules **2** to be installed increases. However, the fastening devices **3** can reduce the time required for the checking work.
[0076] Further, the pair of claws **44** has a restoring force. The distance W**1** between the pair of claws **44** is smaller than the width W of the base portion **60**. Therefore, the pair of claws **44** can regulate the pressing member **23** from moving to the second state P**2** after the pressing member **23** is in the first state P**1**, and thus the pressing member **23** can be prevented from coming off the support member **22** when the nut **25** is loosened or when replacing the bolt **24** and the nut **25**.
[0077] Further, since the pair of claws **44** can be formed together when the support member **22** is formed by press working, the support member **22** can be manufactured easily and inexpensively.
[0078] Further, the fastening device **3** fastens the support member **22** to the base member **20** together with the pressing member **23** by using the bolt **24** and the nut **25**. Therefore, the support member **22** and the pressing member **23** can be prevented from being displaced, and thus the efficiency of the fastening operation of the photovoltaic module **2** can be improved.
[0079] Furthermore, in the fastening device **3**, the base member **20** is provided between the two photovoltaic modules **2**, and the first regulating portion **62** abuts on the side surface portions **14** of the photovoltaic modules **2** on the ridge side and the eaves side, thereby regulating the movement of the photovoltaic modules **2** in the third direction V**3**. In the fastening device **3**, the base member **20** is provided between two photovoltaic modules **2**, and the second regulating portion **63** abuts on the upper surface portions **13** of the photovoltaic modules **2** on the ridge side and the eaves side or are separated from the upper surface portions **13** with a gap therebetween, and thus upward movement of the photovoltaic modules **2** is regulated. Therefore, the movement of the two photovoltaic modules **2** can be regulated by one fastening device **3**.
[0080] An upper surface of the base portion **30** of the base member **20** is formed into a flat surface parallel to the roof board **202** in a state where the base member **20** is fastened to the roofing member **201**. Therefore, the photovoltaic module **2** can be fastened in parallel with the roof board **202**.
[0081] Furthermore, the base member **20** can move the support member **22** and the pressing member **23** in the temporarily fixed state along the groove **35**, and thus the positions of the support member **22** and the pressing member **23** can be adjusted during the fastening operation of the

photovoltaic module **2**. Without the need to remove the base member **20** during the adjustment of the positions of the support member **22** and the pressing member **23**, the efficiency in the operation of fastening the photovoltaic modules **2** can be improved.
[0082] Moreover, the base member **20** is configured such that the upper surface of the base portion **30** becomes parallel to the roof board **202** owing to the length of the vertical wall portion **34** in the vertical direction. This simplifies the configuration of the base member **20**.
[0083] The groove **35** includes the first portion **37** through which the head portion **24***a* can be inserted. Therefore, the bolt **24** can be replaced without detaching the base member **20** from the roof **200**. The base member **20** further includes a regulating portion **36** that regulates rotation of the bolt **24**. Therefore, the efficiency in fastening of the bolt **24** and the nut **25** can be improved. Furthermore, the regulating portion **36** includes a pair of vertical wall portions **36***a* formed at a pair of edges of the groove **35**. Therefore, the configuration of the regulating portion **36** can be simplified.
[0084] In the support member **22**, a first protrusion **48***a* to become electrically conductive to a base part of the frame **11** of the ridge-side photovoltaic module **2** and a second protrusion **48***b* to become electrically conductive to the base part of the frame **11** of the eaves-side photovoltaic module **2** are formed on the mounting portion **48** for mounting the eaves-side and ridge-side photovoltaic modules **2**. Thus, the conduction between each photovoltaic module **2** and the support member **22** can be achieved by simply arranging each photovoltaic module **2** on the support member **22**, and the fastening device **3** can easily perform the conduction setting operation.
[0085] The first protrusion **48***a* and the second protrusion **48***b* formed on the support member **22** can be formed by bending a part of the mounting portion **48** when the support member **22** is subjected to press working. Therefore, the support member **22** can be manufactured easily and inexpensively.
[0086] Further, since the first protrusion **48***a* is electrically conductive to the base part of the side surface portion **14** of the frame **11** of the ridge-side photovoltaic module **2**, when the photovoltaic module **2** is inserted between the support member **22** and the pressing member **23**, the first protrusion **48***a* can be prevented from obstructing the insertion of the photovoltaic module **2**.
[0087] The pressing member **23** has a regulating protrusion **62***a* in the first regulating portion **62**, which prevents the photovoltaic module **2** in the oblique posture from being inserted in the second posture P**2**. Thus, the fastening device **3** can prevent the photovoltaic module **2** from being erroneously disposed between the support member **22** and the pressing member **23** in a state where the bolt **24** and the nut **25** are not engaged.
[0088] As described above, according to the fastening device **3** having the fastening tool **6** according to the embodiment, it is possible to easily check that the fastening tool **6** has been forgotten to be fastened.
[0089] The present invention is not limited to the foregoing embodiment. In the above-described example, the pair of claws **44** are integrally formed with the pair of vertical wall portions **47** of the support member **22** by bending or the like, but it is not limited thereto. For example, the pair of claws **44** may be formed separately from the pair of vertical wall portions **47** and fastened to the vertical wall portions **47** by welding, fastening with a fastening member such as a screw or a rivet, bonding with an adhesive, etc., or the like after the support member **22** is molded. In addition, an example in which the pair of claws **44** are elastically deformed when pressed by the pressing member **23**, and are restored after the pressing member **23** passes over the pair of claws **44** has been described, but it is not limited thereto. For example, in the case where the pair of claws **44** are pressed by the pressing member **23**, the pair of claws **44** may be configured to plastically deform or be damaged and not be restored.
[0090] That is, the present invention is not limited to the above-described embodiments, and various modifications can be made without departing from the scope of the present invention in the implementation stage. The embodiments may be appropriately implemented in combination, and in

this case, combined effects can be obtained. Further, the above-described embodiments include various inventions, and various inventions can be extracted by combinations selected from a plurality of disclosed constituent features. For example, even if some of the constituent features are deleted from all the constituent features described in the embodiments, the configuration from which the constituent features are deleted can be extracted as an invention as long as the problem can be solved and the effects can be obtained.

Claims

1. A fastening tool to mount a plate-shaped photovoltaic module to a mounting target object, comprising: a support member including a pair of vertical wall portions and a pair of mounting portions formed at upper ends of the pair of vertical wall portions and configured to mount the photovoltaic module thereon, the support member being arranged on the mounting target object; a pressing member including a base portion fastened to the mounting target object by a fastening member via the support member, and a regulating portion provided at the base portion and configured to regulate movement of the photovoltaic module in a direction away from the mounting target object, the pressing member being arranged between the pair of vertical wall portions; and a pair of claws configured to deform by an external force, the pair of claws being formed at the pair of vertical wall portions, being tilted with respect to the vertical wall portions, and having a distance between tips smaller than a width of the base portion, wherein the pair of claws are configured to, prior to the fastening of the base portion to the mounting target object by the fastening member, abut on the base portion, and tilt the pressing member with respect to a posture of the regulating portion regulating movement of the photovoltaic module.
2. The fastening tool according to claim 1, wherein the claws are configured to elastically deform by the external force.
3. The fastening tool according to claim 1, wherein the regulating portion includes a first regulating portion and a second regulating portion, the first regulating portion being provided at the base portion, standing upward from the base portion, and facing a side surface of the photovoltaic module, the second regulating portion being provided at the first regulating portion and facing a main surface of the photovoltaic module on a side opposite to the mounting target object.
4. The fastening tool according to claim 3, wherein the first regulating portion is arranged between two adjacent photovoltaic modules, and the second regulating portion includes a third regulating portion facing one of the two photovoltaic modules and a fourth regulating portion facing another of the two photovoltaic modules.
5. The fastening tool according to claim 4, wherein the fastening tool includes a first protrusion formed at the mounting portion and electrically conductive to a side surface portion of one photovoltaic module of the two photovoltaic modules, and a second protrusion electrically conductive to a lower surface portion of the other photovoltaic module of the two photovoltaic modules.
6. The fastening tool according to claim 5, wherein the first protrusion and the second protrusion are formed by a part of the mounting portion.
7. A fastening tool to mount two plate-shaped adjacent photovoltaic modules to a mounting target object, comprising: a support member including a pair of vertical wall portions and a pair of mounting portions formed at upper ends of the pair of vertical wall portions and configured to mount the photovoltaic module thereon, the support member being arranged on the mounting target object; a pressing member including a base portion fastened to the mounting target object by a fastening member via the support member, and a regulating portion provided at the base portion and configured to regulate movement of the photovoltaic module in a direction away from the mounting target object, the pressing member being arranged between the pair of vertical wall portions; a first protrusion formed at the mounting portion and electrically conductive to a side

surface portion of one photovoltaic module of the two photovoltaic modules; and a second protrusion formed at the mounting portion and electrically conductive to a lower surface portion of the other photovoltaic module of the two photovoltaic modules.

8. The fastening tool according to claim 7, wherein the first protrusion and the second protrusion are formed by a part of the mounting portion.

9. A fastening device, comprising: the fastening tool according to claim 1; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.

10. A fastening device, comprising: the fastening tool according to claim 2; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation location of the photovoltaic module.

11. A fastening device, comprising: the fastening tool according to claim 3; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation location of the photovoltaic module.

12. A fastening device, comprising: the fastening tool according to claim 4; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.

13. A fastening device, comprising: the fastening tool according to claim 5; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.

14. A fastening device, comprising: the fastening tool according to claim 6; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.

15. A fastening device, comprising: the fastening tool according to claim 7; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.

16. A fastening device, comprising: the fastening tool according to claim **8**; and at least one of the fastening member and the mounting target object, which is a base member fastened to an installation position of the photovoltaic module.
